50 DevOps Interview Questions

1. General DevOps Concepts

1. What is DevOps, and why is it important?
2. How does DevOps differ from Agile and traditional IT operations?
3. What are the key benefits of DevOps?
4. What are some key DevOps tools you have worked with?
5. Explain the DevOps lifecycle.

2. Linux & Shell Scripting

6. What are some common Linux commands used in DevOps?
7. How do you schedule a cron job in Linux?
8. What is the difference between grep, sed, and awk?
9. How do you check system performance in Linux?
10. How would you write a Bash script to automate a task?

3. Version Control (Git & GitHub/GitLab)

11. What is Git, and why is it used in DevOps?
12. Explain the difference between Git merge and Git rebase.
13. What is a Git branching strategy?

14. How do you resolve merge conflicts in Git?
15. Explain the purpose of a Git commit, push, pull, and fetch.

4. CI/CD (Jenkins, GitHub Actions, GitLab CI/CD)

16. What is Continuous Integration (CI), and why is it important?
17. What is Continuous Deployment (CD), and how does it differ from Continuous Delivery?
18. How would you set up a CI/CD pipeline?
19. What are some common CI/CD tools, and how do they compare?
20. How do you handle rollback in a CI/CD pipeline?

5. Cloud Computing (AWS, Azure, GCP)

21. What are the benefits of cloud computing in DevOps?
22. What are the main services provided by AWS for DevOps?
23. What is the difference between EC2, ECS, and EKS in AWS?
24. What is IAM, and how does it help in security?
25. How would you deploy an application in the cloud?

6. Containers & Orchestration (Docker & Kubernetes)

26. What is Docker, and how does it work?
27. Explain the difference between a Docker image and a Docker container.

28. What is Kubernetes, and why is it used?
29. What is a Kubernetes Pod, and how does it work?
30. What is the difference between a Deployment and a StatefulSet in Kubernetes?

7. Infrastructure as Code (Terraform, Ansible)

31. What is Infrastructure as Code (IaC), and why is it important?
32. How does Terraform differ from Ansible?
33. What are Terraform modules, and how do they work?
34. Explain the difference between terraform plan and terraform apply.
35. How would you manage secrets in Terraform?

8. Monitoring & Logging (Prometheus, Grafana, ELK Stack)

36. What is monitoring in DevOps, and why is it necessary?
37. What is the difference between Prometheus and Grafana?
38. How would you set up an alert in Prometheus?
39. What is the ELK Stack, and what is it used for?
40. How do you debug performance issues in a production system?

9. Security in DevOps (DevSecOps)

41. What is DevSecOps, and how does it integrate into DevOps?

42. What are some best practices for securing CI/CD pipelines?
43. What tools do you use for security scanning in DevOps?
44. How would you handle secrets management in DevOps?
45. How do you ensure compliance with security policies in DevOps workflows?

10. Troubleshooting & Real-World Scenarios

46. How would you handle a failing deployment?
47. What steps would you take if a Kubernetes pod is stuck in a crash loop?
48. How do you troubleshoot high CPU usage in a cloud environment?
49. How would you optimize a slow-running CI/CD pipeline?
50. Tell me about a time when you solved a major DevOps challenge.

These questions will help candidates prepare for real-world DevOps interviews.